Project 2: Markets Quantitative Analysis

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Target

The following task requires the candidate to demonstrate their ability to apply advanced quantitative finance techniques to the valuation of coffee derivatives. Using the provided market data, the individual is asked to address three key areas:

- Cost of Carry Model – The candidate should calculate the theoretical futures price of coffee by incorporating the spot value, risk-free rate, storage costs, and time to maturity. This ensures that the futures contract reflects the true cost of holding the commodity.
- Black-Scholes Model – The candidate should apply the Black-Scholes framework to price a coffee call option, making use of the given strike price, volatility, and time to maturity. This highlights how option pricing depends on both market conditions and stochastic parameters.
- Managing Risk – The candidate should perform simulations of coffee price paths under a geometric Brownian motion to estimate the distribution of option payoffs. This allows for the assessment of risk and uncertainty in option valuation, while also testing sensitivity to shocks in supply, demand, weather, or geopolitical factors.

By addressing these elements, the candidate demonstrates technical expertise in futures and options pricing, as well as the ability to incorporate real-world risk drivers into robust financial models.

Script

The following Python script calculates the fair value of a coffee futures contract using the cost of carry approach.

Cost of Carry model. It defines the spot price, risk-free rate, storage cost, and time to maturity as the key inputs driving the model.

```
import numpy as np

# Given values
S_t = 1.20  # Spot price in dollars
r = 0.02  # Risk-free rate (2%)
d = 0.01  # Storage cost (1%)
T = 0.5  # Time to maturity in years
```

Based on these parameters, the script applies the exponential cost of carry formula to obtain the futures price. Finally, a graph is generated to illustrate the relationship between spot and futures prices, highlighting both the given spot level and the corresponding theoretical futures value.

```
# Calculating futures price
F_t = S_t * np.exp((r + d) * T)
print(f"The fair price of the coffee futures contract is ${F_t:.3
    f} per pound.")
```

Black School model. The Black-Scholes model is used to calculate the fair price of a call option on coffee futures. The process begins by defining the key parameters: the current spot price, strike price, risk-free rate, volatility, and time to maturity.

```
from scipy.stats import norm
import numpy as np

# Given values
S_0 = 1.20  # Spot price in dollars
X = 1.25  # Strike price in dollars
r = 0.02  # Risk-free rate (2%)
T = 0.5  # Time to maturity in years
sigma = 0.25  # Volatility (25%)
```

With These inputs, the model computes $d1$ and $d2$, which incorporate both the expected growth of the asset and the uncertainty from volatility. These terms are then evaluated using the cumulative normal distribution to capture the probability-adjusted outcomes of the option expiring in the money.

```
# Calculating d1 and d2
d1 = (np.log(S_0 / X) + (r + 0.5 * sigma ** 2) * T) / (sigma * np
    .sqrt(T))
d2 = d1 - sigma * np.sqrt(T)
```

Finally, the call option price is obtained by discounting the strike component at the risk-free rate and combining it with the spot price component, resulting in a fair value that reflects both market conditions and the stochastic nature of asset prices.

```
# Calculating call option price using Black-Scholes formula
C = S_0 * norm.cdf(d1) - X * np.exp(-r * T) * norm.cdf(d2)
print(f"The price of the call option is ${C:.3f}.")
```

Monte Carlo simulation. The Monte Carlo simulation code models the future price of coffee by generating thousands of potential price paths under uncertainty. It begins by defining the key parameters, including the spot price, risk-free rate, volatility, and time to maturity, before discretizing the period into daily steps.

```
import numpy as np
import matplotlib.pyplot as plt

# Simulation parameters
S_0 = 1.20  # Spot price in dollars
r = 0.02    # Risk-free rate (2%)
sigma = 0.25  # Volatility (25%)
T = 0.5     # Time to maturity in years
num_simulations = 10000  # Number of simulations
num_steps = 252  # Number of steps (daily)
dt = T / num_steps # Time increment
```

Random shocks are introduced through standard normal variables, allowing the process to follow a Geometric Brownian Motion that captures both drift and volatility effects.

```
# Simulating price paths
np.random.seed(42)  # For reproducibility
price_paths = np.zeros((num_steps, num_simulations))
price_paths[0] = S_0

for t in range(1, num_steps):
    z = np.random.standard_normal(num_simulations)
    price_paths[t] = price_paths[t-1] * np.exp((r - 0.5*sigma**2)
        ↪ *dt +
                                                sigma*np.sqrt(dt)
                                                    ↪ * z)
```

By iteratively simulating these paths, the code produces a distribution of possible terminal prices, from which the average value at maturity is calculated.

```
# Calculating the average simulated price at maturity
average_simulated_price = np.mean(price_paths[-1])
print(f"The average simulated price of the coffee futures 
    ↪ contract at maturity is ${average_simulated_price:.3f}.")
```

Finally, the results can be visualized through a subset of simulated trajectories, highlighting the inherent variability in commodity prices and the expected fair price under risk-neutral assumptions.

```
plt.figure(figsize=(10,6))

# Plot a subset of paths (e.g., 20) for clarity
for i in range(20):
    plt.plot(price_paths[:, i], lw=1)

# Add horizontal line for average price at maturity
plt.axhline(average_simulated_price, color="red", linestyle="--",
            label=f"Average Price at Maturity = ${
                ↪ average_simulated_price:.3f}")

plt.title("Monte Carlo Simulation of Coffee Futures Prices")
plt.xlabel("Days")
plt.ylabel("Price ($/lb)")
plt.legend()
plt.grid(True)
plt.show()
```

Results

Based on the quantitative models, the following outcomes were obtained:

1. The fair price of the coffee futures contract is $1.218 per pound.
2. The price of the call option is $0.068.
3. The average simulated price of the coffee futures contract at maturity is $1.210.

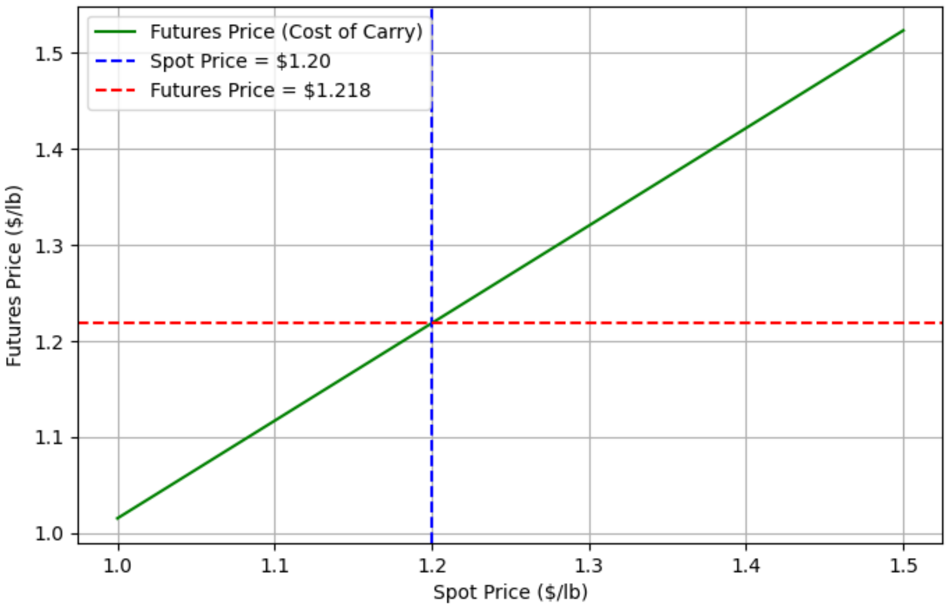

Figure 1: Coffee Futures Pricing using Cost of Carry Model — resultado.

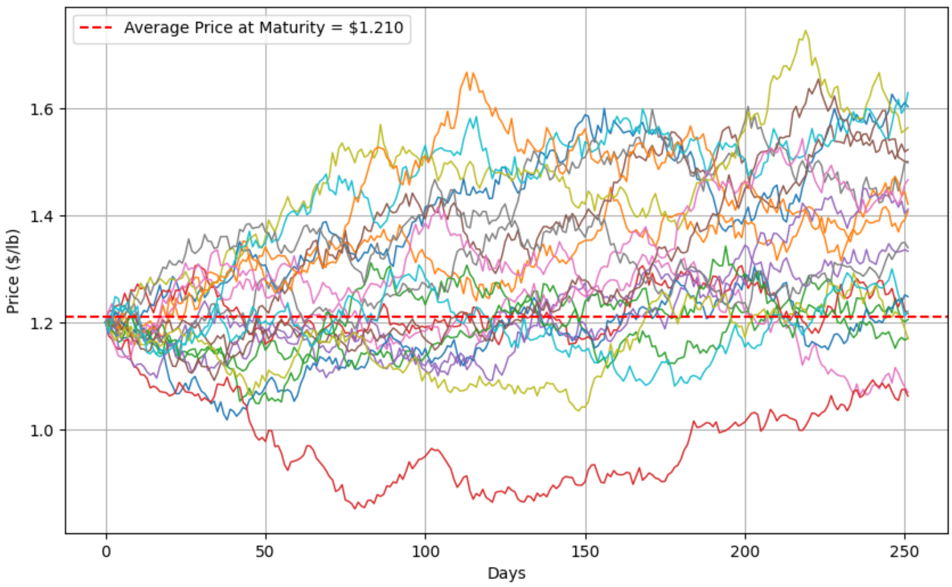

Figure 2: Monte Carlo Simulation of Coffee Futures Prices — resultado.

Conclusion

From the analysis of the structured product, the following conclusions can be drawn:

1. The Cost of Carry model effectively establishes a fair benchmark for futures pricing by incorporating spot value, storage costs, and the risk-free rate.

2. The Black-Scholes model highlights how volatility, strike price, and time to maturity directly influence option valuation, providing a transparent pricing framework.

3. The Monte Carlo simulation demonstrates market uncertainty by generating a distribution of possible terminal prices, capturing both expected value and risk.

4. Together, these models illustrate how combining deterministic and stochastic approaches provides a more comprehensive understanding of derivative pricing and risk management in commodity markets.